

CHAPTER 5

Description of Billiard systems



Finally we start discussing a very widely studied class of model systems: *billiards*. In general they consist of a material point confined in some region of $\mathbb{R}^n$ or $\mathbb{T}^n$ with piecewise smooth boundaries;¹ in the simplest situation such a point moves with constant velocity until it reaches the boundary, and at the boundary it undergoes an elastic reflection. Such models include, e.g., a system of n hard spheres that interacts via elastic collisions (see section 5.6); the importance of such a system as a basic model in statistical mechanics can be hardly overestimated.

These systems are conceptually extremely simple, yet they have an unpleasant feature: they lack smoothness. As we will see in the following there are three main types of non-smoothness: a) tangent collisions; b) collision with a corner; c) accumulation of infinitely many collisions in a finite time. Due to such pathologies these models, in spite of their simplicity, may present some incredibly annoying complications in their treatment.

Let $\mathcal{B}$ be the region in which the point is allowed to move and suppose that $\partial\mathcal{B}$ is a finite union of smooth manifolds with boundary. Clearly the motion can be seen as a flow ϕ_t on the unitary tangent bundle of $\mathcal{B}$ (in fact, given the initial position and the initial velocity the following motion is uniquely determined, moreover the modulus of the velocity will be constant through the motion, so it can be assumed equal to one without loss of generality).²

It can be checked directly that the flow is symplectic (Hamiltonian) in $\tilde{X} := \mathcal{B} \times \mathbb{R}^n$ (see problem Problem 5.6). So, calling m the measure induced

¹Although one can easily consider billiards in a region of a Riemannian manifold with piecewise smooth boundaries, in this case the motion in the interior is just the geodesic flow; see [17] for such a general setting and section ?? for a similar generalization.

²A little thought will convince the reader that two motions with initial velocities that differ only in modulus will be exactly the same apart from the fact that they are run at different speeds.

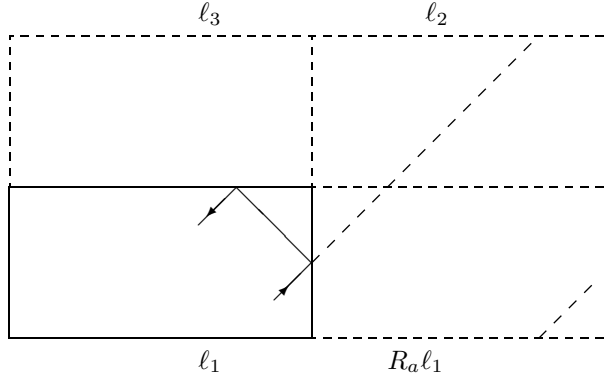


Figure 5.1: The trajectory beyond the mirror

by Lebesgue on $X \equiv \mathcal{B} \times \{v \in \mathbb{R}^n \mid \|v\| = 1\}$, (X, ϕ_t, m) is a smooth flow with collisions (crf. Examples 3.7.1).

5.0.1 Examples

5.0.1.a Polygonal Billiards

The name is self-explanatory: the domain $\mathcal{B}$ is a polygon. The simplest case is probably a rectangle: $\mathcal{B} = [0, a] \times [0, b] \subset \mathbb{R}^2$. Although the notion is fairly trivial, to study it we will employ a neat trick that has many other applications (e.g., see 5.4.1). Consider a trajectory $x + vt$ that reaches the wall $\ell_1 := \{(a, y)\}$. The law of reflection states that, if $v = (v_1, v_2)$, the reflected velocity is $(-v_1, v_2)$. Now define the map $R_a(x, y) = (2a - x, y)$. This is a reflection ($R_a^2 = \text{identity}$) with respect to the wall $\{(a, y)\}$. Remark that $R_a\mathcal{B} = [a, 2a] \times [0, b]$, moreover $DR_a(-v_1, v_2) = v$. This means that, in the reflected box $R_a\mathcal{B}$, the reflected velocity is equal to the velocity before reflection.

The above algebraic discussion corresponds to a very intuitive geometrical fact: if the wall is a mirror, then the trajectory in the mirror is the continuation of the trajectory before collision (see figure 5.1).

After noticing this it is quite clear that one can understand better the trajectory in the “universal covering” of the box obtained by reflecting the box repeatedly with respect to its walls. In this covering the trajectory is simply a straight line and the trajectory in the original box is obtained by undoing the reflections (for the more mathematical inclined let us say that the plane is covered by equal boxes that are identified via reflections, see Problem 5.1). It is then obvious that, given the original velocity v only four velocities are possible: $(\pm v_1, \pm v_2)$. In fact, if we identify the opposite sides in figure 5.1 (that is, side ℓ_3 with side ℓ_1 , side ℓ_2 with $R_a\ell_1$ and so forth) we obtain exactly a flat torus with sides twice as long

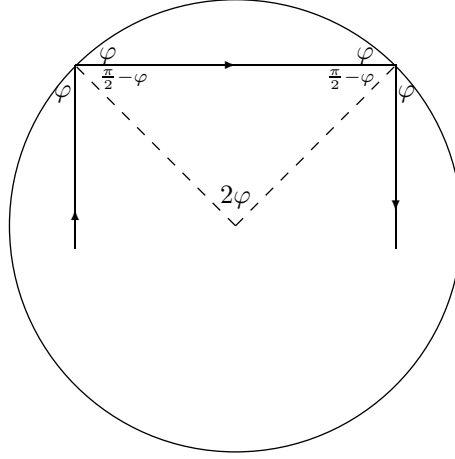


Figure 5.2: Trajectory in a Circular billiard

as the ones of the original rectangle. In addition, the motion on such a torus corresponds precisely to the flow at unit speed in direction v . In other words the motion is equivalent to the rigid translations (geodesic flow) on the associated torus.

Accordingly, the motion is ergodic only if $\frac{v_1 b}{v_2 a}$ is irrational (see).

5.0.1.b Circular Billiards

In this case $\mathcal{B}$ is a disk of radius r . For convenience, let us center it at the origin of a Cartesian coordinate frame. Let us consider a point that has just collided with the boundary at the position $rn(\theta) := r(\cos \theta, \sin \theta)$, where θ is the angle with the x axis counted counterclockwise, and has velocity $v(\theta - \varphi) := (-\sin(\theta - \varphi), \cos(\theta - \varphi))$, which means that the velocity forms an angle φ with the tangent at the collision point. Accordingly, the trajectory will move along the cord of length $2r \sin \varphi$ and collide with the angle $\pi - \varphi$ which, after reflection, will be φ again.

This phenomena is nothing else than the conservation of the angular momentum (for the mechanical inclined) or of the Clairiot integral (for the differential geometers).

All the above implies that, if $\frac{\varphi}{\pi} \in \mathbb{Q}$, then the motion will be periodic, otherwise the collision point will perform an irrational rotation on the boundary. In fact, let us choose as coordinates the distance τ from the last collision point computed along the trajectory; the distance s , computed along the circumference, of the last collision point from a fixed point on the circumference; and the angle φ .

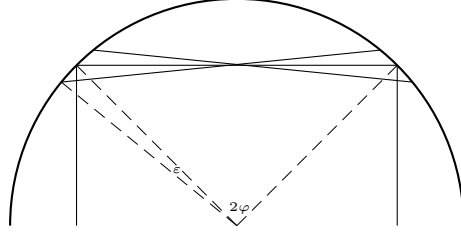


Figure 5.3: Focusing trajectories in a circular billiard

Then the phase space is

$$X = \{(\tau, s, \varphi) \in [0, r] \times S^1 \times [0, \pi] \mid 0 \leq \tau \leq 2r \sin \varphi\}$$

and the flow is nothing else than a suspension flow (see section 1.3) with ceiling function $2r \sin \varphi$ constructed on the map T defined by

$$T(s, \varphi) = (s + 2\varphi, \varphi).$$

At the same time the middle point of the chords between two consecutive collisions will describe an irrational rotation on the circle of radius $r(1 - \cos \varphi)$. Note that this means that the trajectory under consideration (if $\varphi/\pi \notin \mathbb{Q}$) covers densely a two dimensional torus in the three dimensional space (see Problem 5.5) and it is ergodic restricted to it. The circle of radius $r(1 - \cos \varphi)$ is called *caustic*; the name derives from optic because if the trajectory is run by a beam of light that is the place with the highest luminosity.³

To understand better the concept of caustic consider a bunch of near by trajectories that all collide with the same angle φ , see Figure 5.3. If we consider two trajectories colliding at s and $s + \varepsilon$, then it is an easy exercise in elementary geometry to compute the length of the segment on the first trajectory between the collision and the point in which the two trajectories intersect. The result is

$$2r \sin \frac{\varepsilon}{2} \cos(\varphi - \frac{\varepsilon}{2}) + \frac{2r \sin \frac{\varepsilon}{2} \sin(\varphi - \frac{\varepsilon}{2}) \cos \varepsilon}{\sin \varepsilon} = r \sin \varphi + \mathcal{O}(\varepsilon^2). \quad (5.0.1)$$

That is the two trajectories meet (focus) at the middle point apart from an error of order ε^2 . Thus a bunch of trajectories nearby to a given one will all focus approximatively on the middle point, the *burning* point indeed.

³In ancient Greek caustic (*καυστικός*) means “that burns”. Of course, that would be an important concept if you want, e.g., burn a Roman ship. (check the Optic of Euclid)

The above examples correspond to very regular motions (“integrable motion”) that is exactly the opposite of what we mean to investigate. Unfortunately, to progress in the direction we are interested in many more technical tools are needed. Yet, before going on with general facts and definitions let us anticipate two concrete examples that will be particularly relevant in the following.

5.1 Sinai Billiard

The simplest example of Sinai billiards (introduced in [79] and studied in [80]) are given when $\mathcal{B} \subset \mathbb{T}^2$. More precisely, given a disk D , centered at the origin and with diameter $r < \frac{1}{2}$, let $\mathcal{B} = \mathbb{T}^2 \setminus D$. Calling $(x, v) \in \mathcal{B} \times \mathbb{R}^2$ the position and the velocity, respectively, the motion is described by a free flow

$$\phi_t(x, y) = (x + vt, v), \quad (5.1.1)$$

provided $\|x + vt\| \geq r$, that is provided the motion does not exist $\mathcal{B}$. When $x \in \partial\mathcal{B} = \partial D$ a collision takes place. Of course, at the collision it must be $\langle x, v \rangle \leq 0$, the velocity points toward D , otherwise the point would not have reached the obstacle D but rather would be flowing away from it. The collision law is, as already said, an elastic collision—namely the total energy and the momentum tangential to the collision plane must be preserved. Thus, calling v_- the velocity before collision and v_+ the velocity after collision, we require

$$\|v_+\| = \|v_-\| ; \quad \langle Jx, v_- \rangle = \langle Jx, v_+ \rangle,$$

where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

so that $\langle Jx, x \rangle = 0$, that is $r^{-1}Jx$ is a unit vector tangent to the disk and oriented counterclockwise. This implies:

$$v_+ = v_- - \frac{2}{r^2} \langle x, v_- \rangle x. \quad (5.1.2)$$

5.1.1 Flow

From the above discussion it is clear that (X, ϕ^t, m) is a smooth flow with collisions, the only property that need to be checked is (3.7.1) (see Problem ???).

Let us call $V(x, v) = (v, 0)$ the vector field generating ϕ^t . A useful fact is the following.

Lemma 5.1.1 *If $w \in \mathcal{T}_\xi X$ and $\langle w, V(\xi) \rangle = 0$, then $\langle d\phi^t w, V(\phi^t(\xi)) \rangle = 0$.*

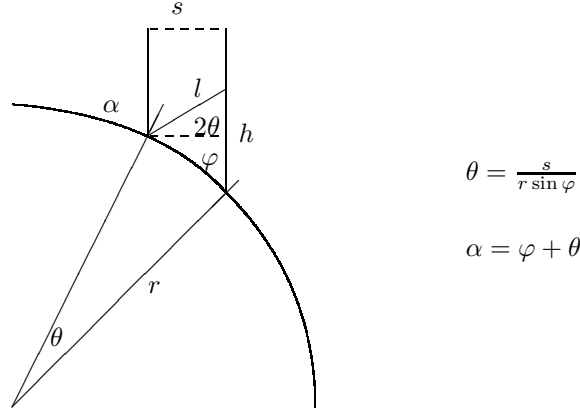


Figure 5.4: Collision

PROOF. If no collision takes place, then the statement it is obvious by equation (5.1.1) and since for each $w = (w_1, w_2) \in \mathcal{T}X$ it must be $\langle w_1, v \rangle = 0$ (just differentiate $\|v\|^2 = 1$). Let us see what happens at collision.

Given the tangent vector $w = (w_1, w_2)$ at the point $\xi \in X$, we can consider the curve $\gamma(s) = \xi + ws$ that generates it ($\gamma'(0) = w$). Suppose that the next collision takes place with an angle φ . If we refer to the Figure 5.4 all we need to compute is the relation between h and l . A bit of geometry shows that

$$h = s \arctan \varphi + \mathcal{O}(s^2); \quad l = \frac{s}{\cos \theta} \arctan \varphi + \mathcal{O}(s^2) = s \arctan \varphi + \mathcal{O}(s^2).$$

Thus, if τ is the collision time of the trajectory starting at ξ and $\tilde{\gamma}(s) = \phi^{\tau+}(\gamma(s))$, we have $\tilde{\gamma}'(0) = d_\xi \phi^{\tau+} w := \tilde{w}$, and, calling v_+ the velocity after reflection, $\langle v_+, \tilde{w} \rangle = 0$, which proves the lemma. $\square$

This means that in this case there is a particularly simple way to quotient out the flow direction (see ???): consider only vectors perpendicular to the flow.

5.1.2 Poincaré map

For many purposes it is useful to view the Sinai billiards as a symplectic map from a two dimensional domain to itself. Such a reduction is obtained via a general technique widely used in dynamical system: a Poincaré section (see 1.2). A Poincaré section consists in introducing some codimension one manifolds in the phase space X and then defining a map from such manifolds to themselves in such a way that to each point is associated its first return to the manifolds (if it exists). Let us be more concrete.

Historically the choice of the section to realize a Poincaré map as been based on $\partial\mathcal{B}$. In our case this consists of the boundary of the disk, that is a circle. Of course, it is also necessary to specify the velocity. Clearly there are two possibilities: one can consider velocities just before collision, which means $\langle x, v \rangle \leq 0$, (this is the Poincaré map from before collision to before collision) or one can consider the velocity just after collision, meaning $\langle x, v \rangle \geq 0$, (that is the Poincaré map from just after collision to just after the next collision). The two choices are equivalent, let us make the second.

If we identify the velocity by the angle φ between the tangent (directed clockwise) to the disk at the collision point and the velocity vector v , then the phase space is $\mathcal{M} = S^1 \times [0, \pi]$.

We can then define a map $T : \mathcal{M} \rightarrow \mathcal{M}$ in the following way: for each $\xi \in \mathcal{M}$, let $T\xi$ be the point just after the next reflection (if such a reflection exists). Note that, if no reflections would occur, almost all the trajectories would fill $\mathbb{T}^2$ densely,⁴ hence T is defined almost everywhere.

It is natural to use as coordinate on the boundary the distance s , computed counterclockwise along the circle, from a given point. If we want to compute the induced invariant measure on the Poincaré section $\mathcal{M}$, according to section ?, we have to introduce the change of coordinates $\Xi : \mathcal{M} \times [0, \delta] \rightarrow X$ defined by

$$\Xi(s, \varphi, t) = (rn(sr^{-1}) + v(sr^{-1} + \varphi)t, sr^{-1} + \varphi - \frac{\pi}{2}),$$

where $n(\theta) = (\cos \theta, \sin \theta)$, $v(\theta) = (\sin \theta, -\cos \theta)$. In this coordinates a point is determined by its collision data (s, φ) and the time t past from the last collision.

A direct computation shows that

$$\begin{aligned} \det \Xi &= \begin{vmatrix} -v(sr^{-1}) + r^{-1}n(sr^{-1} + \varphi)t & n(sr^{-1} + \varphi) & v(sr^{-1} + \varphi) \\ r^{-1} & 1 & 0 \end{vmatrix} \\ &= \begin{vmatrix} -v(sr^{-1}) & n(sr^{-1} + \varphi) & v(sr^{-1} + \varphi) \\ 0 & 1 & 0 \end{vmatrix} \\ &= \begin{vmatrix} -v(sr^{-1}) & v(sr^{-1} + \varphi) \end{vmatrix} = \sin \varphi. \end{aligned}$$

So, given a set $A \subset \mathcal{M}$, calling $A_\varepsilon = \cup_{t=0}^\varepsilon \phi^t A$,

$$\mu(A) := \frac{1}{\varepsilon} m(A_\varepsilon) = \frac{1}{\varepsilon} \int_{A_\varepsilon} |\det(\Xi)| ds d\varphi dt = \int_A \sin \varphi ds d\varphi.$$

Thus $d\mu = \sin \varphi ds d\varphi$ and $(\mathcal{M}, T, \mu)$ is a Dynamical Systems.

It is interesting to notice that μ becomes degenerate for $\varphi \in \{0, \pi\}$, which correspond to tangent collisions. Another annoying feature of the above choice

⁴Since for almost all velocities v we would have an irrational translation on $\mathbb{T}^2$, see Problem 5..

is that some trajectories never meet the boundary of the disk (for example consider the initial condition $x = (1, 0)$, $v = (0, 1)$) and other will travel an arbitrarily long time before the next collision.⁵ These facts, although not catastrophic, may look unpleasant to someone. It is therefore relevant to notice that there are several other possible choices for the Poincaré section, each one with its own advantages and disadvantages. Let us see a couple of them.

Consider the fundamental domain $Q = [-\frac{1}{2}, \frac{1}{2}]^2$ of $\mathbb{T}^2$, choose ∂Q as the basis of the Poincaré section. Of course ∂Q it is not a smooth manifold (it consists of four lines). This problem is easily solved by extending the concept of Poincaré section to the case in which the section Σ is a finite (or even countable) union of smooth manifolds; a quick look at section ?? will convince the reader that this generalization is indeed immediate. This section has the advantage of a simple structure, that there is a maximal time from Σ to itself, yet it does not solve the problem of the degeneration of the measure. Here as well we have problems with the trajectories that meet the section at very small angles.

To overcome such a problem one can choose the section $\Sigma \times [\delta, \pi - \delta]$. It is easy to see that if δ is chosen small enough then the only effect is to skip at most one crossing of the boundary Σ (see Problem 5.).

We will keep using the relation between the two dynamical systems (X, m, ϕ_t) and $(\mathcal{M}, \mu, T)$. In particular it is convenient to define the cone family on all TX instead that only on $\mathcal{TM}$. We will see that an invariant cone family on TX induces an invariant cone family on $\mathcal{TM}$.

5.1.3 Singularity manifolds

In this subsection we will study more precisely the singularities of the system and we will verify that they belong to the general setting developed in 3.6. We will consider two different Poincaré sections to give the reader a more complete idea of the situation.

Let us start with the classical section *just after collision*. As already mentioned the phase space is $\mathcal{M} = S^1 \times [0, \pi]$. Clearly the only singularities of the map correspond to coordinates where the next collision is a tangent one. To analyze such a pathology it is more convenient to look at the billiard on the universal covering of the torus. In such a covering the obstacles will form a lattice and the particles move along a straight line between collisions (see figure 5.5).⁶

⁵This property is called *infinite horizon*. We will discuss it further in the sequel, see ???.

⁶This trick is very similar to the one used at the beginning of the chapter to discuss rectangular billiards, only now we take advantage of the periodicity of the torus rather than the invariant properties of the domain with respect to the reflections.

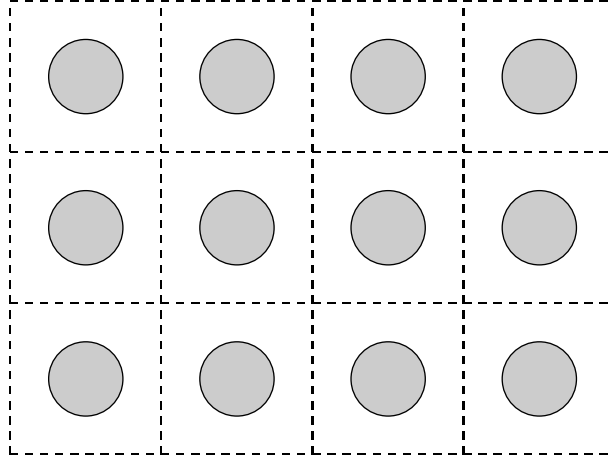


Figure 5.5: Universal covering of a Sinai billiard

The particle with coordinates (s, φ) , just after collision, will move in the direction $v(r^{-1}s + \varphi)$ with unit speed.⁷ Hence, if $C \in \mathbb{R}^2$ is the coordinate of the center of the obstacle with which the next collision will take place, the condition for a tangent collision reads

$$rn(r^{-1}s) + tv(r^{-1}s + \varphi) = C \pm rn(r^{-1}s + \varphi). \quad (5.1.3)$$

Where $t = t(s, \varphi)$ is the collision time. From (5.1.3) we can immediately extract two interesting informations multiplying it by $n(r^{-1}s + \varphi)$ and $v(r^{-1}s + \varphi)$ respectively

$$\begin{aligned} \langle C, v(r^{-1}s + \varphi) \rangle &= t + r \sin \varphi > 0 \\ F(s, \varphi) &:= \langle C, n(r^{-1}s + \varphi) \rangle - r \cos \varphi \pm r = 0 \end{aligned}$$

Taking the derivative of F with respect to φ we get

$$-r \sin \varphi + \langle C, v(r^{-1}s + \varphi) \rangle = t > 0,$$

thus we can apply the implicit function theorem ([1]) and conclude that the manifold corresponding to this discontinuity can be represented as the graph of a function $\varphi(s)$. In addition,

$$\frac{d\varphi}{ds} = -\left(\frac{1}{r} + \frac{\sin \varphi}{t}\right) < 0. \quad (5.1.4)$$

Since there are infinitely many obstacles with which the next collision can take place, there must be countably many discontinuity line (some of them are schematically represented in figure 5.6)

⁷Remember the convention $n(\theta) := (\cos \theta, \sin \theta)$ and $v(\theta) := (\sin \theta, -\cos \theta)$.

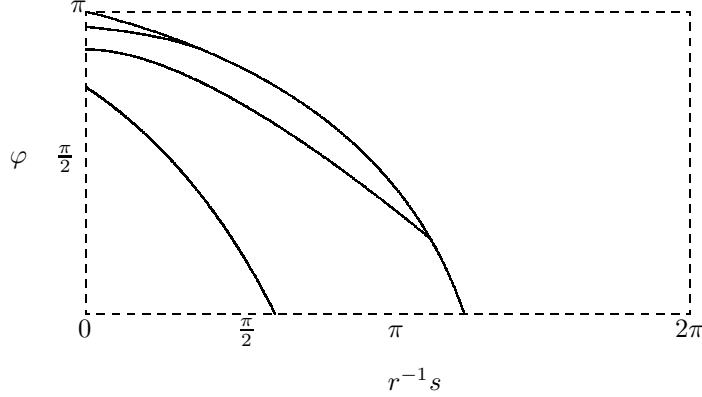


Figure 5.6: Few discontinuity lines in the Poicaré map

Next let us consider a different Poincaré section. In particular we would like to avoid to have a degenerate metric and symplectic form at the boundary of the section. Instead of using one of the proposal advanced in the previous section let us just consider the boundary of the obstacle together with all the angles in the interval $[\delta, \pi - \delta]$, for some sufficiently small δ . In this case we have two types of discontinuities: the tangencies and the preimages of the boundary of the section (before they where the same thing). We will call this type of sections *regularized boundary section*.⁸

To analyze the preimages of the boundary of the section one can proceed in analogy with what we have done before, equation (5.1.3) in this case becomes

$$rn(r^{-1}s) + tv(r^{-1}s + \varphi) = C \pm rn(r^{-1}s + \varphi \pm \delta). \quad (5.1.5)$$

From (5.1.5) we obtain

$$\frac{d\varphi}{ds} = -\left(\frac{1}{r} + \frac{\sin \varphi}{t + r \sin \delta}\right) < 0. \quad (5.1.6)$$

Clearly the map is smooth up to, and including, this type of discontinuity, not so for the tangencies. In fact, it is easy to verify that the map is continuous crossing a tangency line (see Problem 7.8) but we will see immediately that it is not differentiable. By the discussion of section 5.3 (see in particular formulae (5.3.1) and (5.3.2)) it follows that if the next collision takes place with an angle $\varphi \notin [\delta, \pi - \delta]$, then calling τ_1 the time up to the tangent collision

⁸The meaning of the word “regularized” should be obvious but see also Problem 5.7.

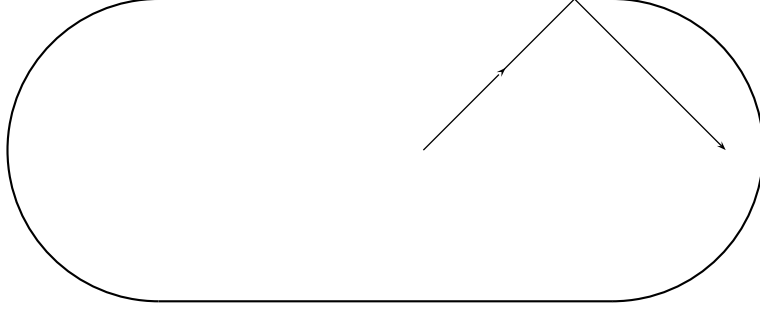


Figure 5.7: Bunimovich Stadium

and τ_2 the time from the tangent collision to the next, we have the formula

$$DT = \begin{pmatrix} 1 + \frac{2\tau_1}{r \sin \varphi} & \frac{2}{r \sin \varphi} \\ \tau_2(1 + \frac{2\tau_1}{r \sin \varphi}) & 1 + \frac{2\tau_2}{r \sin \varphi} \end{pmatrix}.$$

Clearly the norm of DT is bounded by a constant times $\frac{1}{\sin \varphi}$ (if in doubt do Problem 5.9). Now, if a point has distance ε from the singularity line, it will land at a distance $\sqrt{\varepsilon}$ from the tangency, which means that there exists a constant $c_t > 0$ such that, calling $\mathcal{S}$ the singularity line and ξ the point

$$|\sin \varphi| \geq c_t \sqrt{d(\xi, \mathcal{S})}.$$

Which means that the Derivative blows up only as a square root getting close to the singularity. By similar considerations it is possible to verify also that the second derivative blows up polinomially (see Problem 5.10).

5.2 Bunimovich Stadium

These billiards have been introduced [15] and first studied [16] by Bunimovich. In this case the table of the billiard is a convex subset of $\mathbb{R}^2$. The simplest, and original, one consists in two half circles joined by two straight lines (see Figure 5.7).

The name “stadium” is due to the shape of the domain $\mathcal{B}$ in which the motion takes place. The only difference is that now the curvature of the boundary is either zero (collisions against the straight segments) or negative (collision against the half circles).

5.2.1 Flow

We have seen that the flow in a square or in a circle is well defined and rather regular. Clearly the only relevant discontinuity in the Bunimovich Billiard arises when a trajectory hits the joining between the circumference and the straight lines.

5.2.2 Poincaré map

Use coordinates introduced for circle and (eventually) mention Lazutkin

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5.3 Collision map and Jacobi fields

To compute, in general, the collision map it is helpful to introduce appropriate coordinates in $\mathcal{TX}$. A very interesting choice is constitute by the *Jacobi fields*.⁹ Let X_- be the set of configurations just before collision. For each $(x, v) \in X \setminus X_-$ there exists $\delta > 0$ such that

$$\phi_t(x, v) = (x + vt, v) \quad 0 \leq t \leq \delta.$$

Let us consider the curve in $\mathcal{X}$

$$\xi(\varepsilon) = (x(\varepsilon), v(\varepsilon)),$$

with $\xi(0) = (x, v)$ and $\|v(\varepsilon)\| = 1$.

For each t such that $\phi_t(\xi(0)) \notin X_-$, let

$$\xi(\varepsilon, t) = (x(\varepsilon, t), v(\varepsilon, t)) = \phi_t(\xi(\varepsilon)).$$

The Jacobi field $J(t)$ is defined by

$$J(t) \equiv \left. \frac{\partial x}{\partial \varepsilon} \right|_{\varepsilon=0}.$$

⁹The Jacobi Fields are a widely used instrument in Riemannian geometry (see [36]) and have an important rôle in the study of Geodetic flows, although we will not insists on this aspect at present. Here they appear in a very simple form.

Note that, since $x(0, t) \notin X_-$, for $s < \delta$

$$\xi(\varepsilon, t + s) = \xi(\varepsilon, t) + (v(\varepsilon, t)s, 0),$$

so

$$J'(t) = \frac{dJ(t)}{dt} = \frac{\partial v(\varepsilon, t)}{\partial \varepsilon} \Big|_{\varepsilon=0}.$$

That is, $(J(t), J'(t)) = d\phi_t \xi'(0)$.

At each point $\xi = (x, v) \in X$ we choose the following base for $\mathcal{T}_\xi X$:¹⁰

$$\eta_0 = (v, 0); \quad \eta_1 = (v^\perp, 0); \quad \eta_2 = (0, v^\perp);$$

where $\|v^\perp\| = 1$, $\langle v, v^\perp \rangle = 0$.

The vector η_0 corresponds to a family of trajectories along the the flow direction and it is clearly invariant; η_1 to a family of parallel trajectories and η_2 to a family of trajectories just after focusing. It is very useful the following graphic representation. We represent a tangent vector by drawing a curve that it is tangent to it. A curve in $\mathcal{T}X$ is given by a base curve that describes the variation of the x coordinate equipped with a direction at each point (specified by an arrow) which show how varies the velocity (see Figure ??).

A direct check shows that each vector η perpendicular to the flow direction will stay so (see Lemma 5.1.1 and Problem 5.), i.e.

$$\langle d\phi_t \eta, (v_t, 0) \rangle = \langle d\phi_t \eta, d\phi_t(v, 0) \rangle = \langle \eta, (v, 0) \rangle = 0.$$

So the free flow is described by

$$d\phi^t \eta_0 = \eta_0; \quad d\phi^t \eta_1 = \eta_1; \quad d\phi^t \eta_2 = \eta_2 + t\eta_1,$$

that is, in the above coordinates

$$d\phi^t = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & t & 1 \end{pmatrix}. \quad (5.3.1)$$

Let us see now what happens at a collision.

Let $x_0 \in \partial\mathcal{B}$ be the collision point and let $\xi_c = (x_0, v)$ be the configuration at the collision. We want to compute $R_\varepsilon := d_{\phi^{-\varepsilon}\xi_c} \phi^{2\varepsilon}$, that is the tangent map from just before to just after the collision. Clearly $R_\varepsilon \eta_0 = \eta_0$. From the Figure 5.8 follows that, if $\gamma(s)$ is the curve associated to η_1 at the point $\phi^{-\varepsilon}\xi_c$,

$$d\phi^{2\varepsilon} \gamma(s) = \left(v_+^\perp \left[s + \varepsilon \frac{2s}{r \sin \varphi} \right], \frac{2s}{r \sin \varphi} \right) + \mathcal{O}(s^2)$$

¹⁰Here $v^\perp = Jv$ with

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

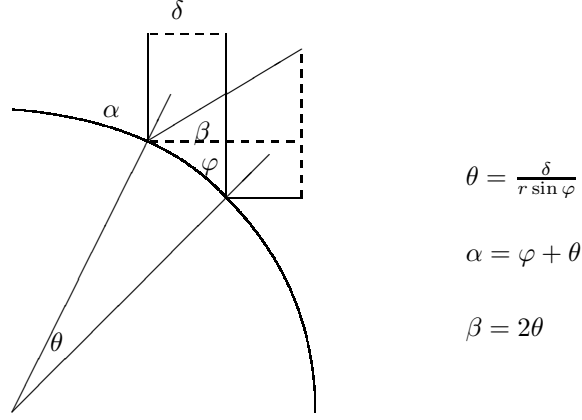


Figure 5.8: Collision

where r is the radius of the osculating circle (that is the circle tangent to the boundary up to second order) which is the inverse of the curvature $K(x_0)$ of the boundary at the collision point.

The above equation means that

$$J(\varepsilon) = \left(1 + \frac{2\varepsilon K(x_0)}{\sin \varphi}\right) v_+^\perp.$$

Accordingly, calling $R = \lim_{\varepsilon \rightarrow 0} R_\varepsilon$ the collision map, we have

$$R\eta_1 = \eta_1 + \frac{2K}{\sin \varphi} \eta_2; \quad R\eta_2 = \eta_2.$$

Hence,

$$DR = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{2K}{\sin \varphi} \\ 0 & 0 & 1 \end{pmatrix}. \quad (5.3.2)$$

The above computations provide the following formula for the derivative of the Poincaré section from the boundary of the obstacle, just after collision, to the boundary of the obstacle just after the next collision

$$DT = \begin{pmatrix} 1 & \frac{2K}{\sin \varphi} \\ \tau & 1 + \frac{2\tau K}{\sin \varphi} \end{pmatrix}, \quad (5.3.3)$$

where τ is the flying time between the two collisions and φ the collision angle.

Formula (5.3.3) is sometimes called *Benettin formula* (e.g., [52], from ???).

the case of two disc in two dimensions and we will comment on the more general case.

5.6.1 Two balls, two dimensions

Let us start by considering the motion of two identical disks of mass one in $\mathbb{T}^2$. Again the motion is linear plus elastic collisions.

Clearly, the disks must have radius r sufficiently small in order to fit in the torus (see Problem 5.), for simplicity we assume $r < \frac{1}{4}$. The phase space is $X = \mathbb{T}^4 \times \mathbb{R}^4$.

If $x_1, x_2 \in \mathbb{T}^2$ are the coordinated of the center of the disks, the velocity changes at collision according to the law

$$\begin{cases} v_1^+ = v_1^- - \langle \eta, v_2^- - v_1^- \rangle (v_2^- - v_1^-) \\ v_2^+ = v_2^- + \langle \eta, v_2^- - v_1^- \rangle (v_2^- - v_1^-) \end{cases} \quad (5.6.1)$$

where η is a unit vector in the direction $x_2 - x_1$.¹⁵

Here there are more integral of motion than in the previous cases: the energy $E = \frac{1}{2}(\|v_1\|^2 + \|v_2\|^2)$ and the total momentum $P = v_1 + v_2$. Thus, if we want to obtain an ergodic systems, we have to reduce the system via the integral of motion. we will then consider that phase spaces

$$X_{E,P} = \{(x_1, x_2, v_1, v_2) \in X \mid \frac{1}{2}(\|v_1\|^2 + \|v_2\|^2) = E; v_1 + v_2 = P\}.$$

Since, in the velocity space, the previous conditions correspond to the intersection between the surface of a four dimensional sphere (S^3) and a two dimensional linear space, the velocity vectors $(v_1 + v_2)$ is contained in a one dimensional circle. Thus, topologically, $X_{E,P} = \mathbb{T}^4 \times S^1$.¹⁶ It is then natural to choose an angle θ as coordinate on S^1 , moreover, since

$$2E = \|v_1\|^2 + \|v_2\|^2 = \frac{1}{2}\|v_1 - v_2\|^2 + \frac{1}{2}\|P\|^2,$$

it is hard to resist setting $v_2 - v_1 = v(\theta)$.¹⁷ Hence,

$$\begin{cases} v_1 = \frac{1}{2}(P - v(\theta)) \\ v_2 = \frac{1}{2}(P + v(\theta)). \end{cases}$$

¹⁵To be precise $x_2 - x_1$ has no meaning since $\mathbb{T}^2$ it is not a linear space. Yet, at collision, the distance between the two disks is $2r$, so the global structure of $\mathbb{T}^2$ is irrelevant and we can safely confuse it with a piece of $\mathbb{R}^2$.

¹⁶Of course, we are considering only the cases $E \neq 0, P \neq 0$.

¹⁷As usual $v(\theta) = (\sin \theta, \cos \theta)$.

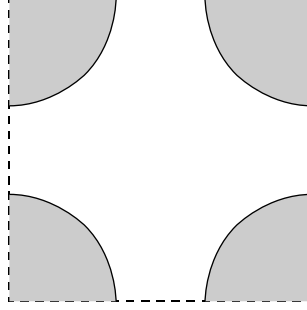


Figure 5.14: Reduced Systems

The free motion is then given by

$$\begin{cases} x_1(t) = x_1(0) + \frac{1}{2}(P - v(\theta))t \\ x_2(t) = x_2(0) + \frac{1}{2}(P + v(\theta))t. \end{cases}$$

Accordingly,

$$\begin{cases} x_1(t) + x_2(t) = x_1(0) + x_2(0) + Pt \pmod{1} \\ x_2(t) - x_1(t) = x_2(0) - x_1(0) + v(\theta)t \pmod{1}. \end{cases}$$

It is then clear the need to introduce the two new variables $Q = x_1 + x_2$ and $\xi = x_2 - x_1$. The variable Q performs a translation on the torus, such a motions is completely understood and we can then disregard it. The only relevant motion is the one in the variables (ξ, θ) . The reduced phase space is then $\cong \mathcal{B} \times S^1$ where $\mathcal{B} = \mathbb{T}^2 \setminus \{\|\xi\| \leq 2r\}$, that is the torus minus a disk of radius $2r$. The domain $\mathcal{B}$ is represented in Figure ?? and, apart the different choice of the fundamental domain, it corresponds exactly to the table of the simplest Sinai billiard.

The free motion corresponds to the free motion of a point as well, while at collision, from (5.6.1), we have

$$v(\theta^+) = v(\theta^-) - 2\left\langle \frac{\xi}{2r}, v(\theta^-) \right\rangle v(\theta^-)$$

that is exactly the elastic reflection from the disk! In conclusion the reduced system is exactly the simplest case of Sinai Billiard already studied in .

5.6.2 Many balls, many dimensions

It is now natural to wonder about n identical hard-balls moving on the d dimensional torus $\mathbb{T}^d$.

Problems

- 5.1** Given a rectangular box $\mathcal{B}$, with its sides labeled by $\{1, 2, 3, 4\}$ and let $R_i(\mathcal{B})$ be the reflection with respect to the side i of the box $\mathcal{B}$.¹⁸ Let R_0 be the identity. Consider $G = \cup_{n=0}^{\infty} \{0, 1, 2, 3, 4\}^n$, if $g \in G$ then we define $R_g(\mathcal{B}) = R_{g_1}(\cdots (R_{g_n}(\mathcal{B})) \cdots)$ and, for each $g^i \in G$, $i \in \{1, 2\}$, $g = g^2 \circ g^1 \in \{0, 1, 2, 3, 4\}^{n_1+n_2}$, is defined by $g_k = g_k^1$ for $k \leq n_1$ and $g_k = g_{n_1-k}^2$, for $k > n_1$. Verify that $R_{g^2}(R_{g^1}(\mathcal{B})) = R_g(\mathcal{B})$. Introduce the equivalence relation $g_1 \sim g_2$ iff $R_{g_1}(\mathcal{B}) = R_{g_2}(\mathcal{B})$. Let $\tilde{G}$ be the collection of the equivalence classes. Verify the $\tilde{G}$ is a commutative group with respect to the operation “ $\circ$ ”. (hint: Note that the geometrical meaning is simply that the final position of the box after a certain number of reflections does not depend on the order of the reflections. Clearly it suffices to check such a property for two reflections.)
- 5.2** Study the motion in a triangular billiard when the angle defining the triangle are all rational multiple of π . (Hint: use reflections again)
- 5.3** Study the motion in an elliptical billiard. (Hint: Verify that there exist an integral of motion.)

Lemma 5.7.1 *If the boundary of a convex billiard has bounded curvature, then there cannot be infinitely many reflections in a finite time.*

PROOF.

□

- 5.4** Show that for the Cardioid there cannot be infinitely many reflections in a finite time.
- 5.5** Verify that the caustics correspond to a two dimensional torus and the motion is ergodic restricted to it. (Hint: ...)
- 5.6** Check that the maps ϕ^t generated by a billiard flow are symplectic. (Hint: It is obvious for the free flow, it remains to check it for the reflections. This can be done by using formulae ???)
- 5.7** Find a change of variable that transforms the symplectic form in a regularized boundary section in the standard symplectic form.
- 5.8** Verify that, in a regularized boundary section, the map is continuous across a singularity line corresponding to a tangency.

¹⁸The label attached to the sides of the reflected box are the one obtained naturally from the old ones.

- 5.9** Prove that, given an $n \times n$ matrix A the norm $\|A\| := \sup_{v \in \mathbb{R}^n} \frac{\|Av\|}{\|v\|}$ where $\|v\| := \sqrt{\sum_n v_n^2}$ satisfies

$$\|A\| \leq \text{constant} \max_{i,j} |A_{ij}|$$

and compute explicitly the optimal constant.

- 5.10** Determine the rate at which the second derivative explode as one gets near to a tangency singularity in the Sinai billiard with one circular obstacle in the torus.
- 5.11** Compute the number of collisions in a convex angle.
- 5.12** Show that in a Sinai billiard on $\mathbb{T}^2$ for which there exists trajectories that spend an arbitrary long time without colliding there must exist trajectories that never collide. (Hint: some continuity...)
- 5.13** Show that in a Sinai billiard on $\mathbb{T}^2$ almost all the trajectories must collide. (Hint: If a trajectory never collides then it performs a translation on the torus. Such translations are always ergodic apart from the case in which the ratio of the components of the velocity is a rational number. These trajectories are only countable.)

- 5.14** Let

$$L_i = \begin{pmatrix} 1 & 0 \\ t_i & 1 \end{pmatrix} \quad \text{and} \quad R_i = \begin{pmatrix} 1 & k_i \\ 0 & 1 \end{pmatrix},$$

for each $u \in \mathbb{R}^+$ write

$$\prod_{i=0}^n R_i L_i \begin{pmatrix} 1 \\ u \end{pmatrix} = \lambda_n \begin{pmatrix} 1 \\ u_n \end{pmatrix}.$$

Show that

$$u_n = k_n + \frac{1}{t_n + \frac{1}{k_{n-1} + \frac{1}{t_{n-1} + \dots + \frac{1}{t_1 + \frac{1}{k_1 + \frac{1}{u}}}}}}$$

And find conditions for the convergence of the continuous fraction. (Hint: see Problem 4.37)

5.15 Study two disk with different masses.

5.16 Prove that the Poincarè map for the Sinai billiard is piecewise Hölder of Hölder exponent $\frac{1}{2}$.

Notes

Do a bit of history, maybe starting from Krylov on. About a bit of physical contest as well? or is it better to fit it in the chapter?

.....

CHAPTER 6

Examples of hyperbolic Billiards



Given the historical relevance of the Billiards described in the previous chapter it is natural to investigate the hyperbolicity of such examples. To this task is devoted the present chapter.

6.1 Sinai Billiard

We define a cone family in the plane perpendicular to the flow direction $(v, 0)$, that is in the plane η_1, η_2 ,¹ this plane is naturally isomorphic to the tangent space of $\mathcal{M}$ (just project along the flow direction) in each non-tangent point. Let us sum up the result obtained in section 5.3. In the case in which no collision takes place we have seen that the parallel family η_1 stay parallel, while the most divergent family (the vector η_2) becomes less divergent (a linear combination of η_1 and η_2 with positive coefficients). This means that the first quadrant (in the η_i coordinates goes into itself but the η_1 side stays put—see Figure ??). Let us study what happens at a reflection. Looking at Figure 5.8 one can see that any divergent family of trajectories will be divergent after collision, and in particular the parallel family will be strictly divergent. To be more precise the η_2 family will go into itself after the collision while the parallel one will be strictly divergent. Again the cone goes strictly inside itself but one side (the η_2 one this time) stays put. Nevertheless, the combination of free motion and reflection clearly sends the cone strictly inside itself.

Since almost all trajectories experience a collision (see Problem 6.1) it follows that Wojtkowski theorem applies and that the Lyapunov exponents are different from zero almost everywhere for the dynamical system $(\mathcal{M}, T, m)$.

¹See section 5.3 for the definition of η_1, η_2 and their properties.

6.2 Bunimovich stadium

The naïve understanding of the previous example is that the obstacle acts as a defocusing mirror and thus makes the trajectories diverge, whereby creating instability. This idea was already present in Krylov's work [58] and was considered the natural mechanism producing hyperbolicity. With this point of view in mind it seems that a table with convex boundaries (in which parallel trajectories are focused after reflections) is unlikely to yield hyperbolic behavior. This impression can be only confirmed by the presence of caustics in smooth convex billiards (see Problem 5. and Notes of chapter 5). It came then as a surprise the discovery by Bunimovich that perturbations of the circle² could be hyperbolic. The main intuition is as follows: although the trajectories after reflection may be focusing, after some time they focus and then become again divergent. Hence, if there is enough time between consecutive collisions, we can still have divergent families going into divergent families (provided we look at the right place). Another, equivalent, point of view is that the instability is measured not just by the change in position but also by the change in velocity, from this point of view a very strong convergence it is not so different from a strong divergence.

To find a new invariant family of cones we have to remember a little the case of circular billiards (see Examples 5.0.1). We have seen that the collision angle is a conserved quantity of the motion. It is then natural to consider, at each point in phase space, the tangent vector η_3 associated to a family of trajectories that, at the next collision with one of the two half circles, will have the same collision angle.

We have defined η_3 in geometrical terms, clearly its expression in terms of η_1 and η_2 changes from point to point (see Problem 6.2). Yet, there are special points (the middle of the cord between two consecutive collision with the same half-circle) in which η_3 coincides with η_2 (this is seen immediately by geometric considerations, see Figure 5.3 and equation (5.0.1)).

Clearly, in a sequence of collisions with the same circumference the vector η_3 is invariant. Also, from the above considerations follows that before collisions η_3 corresponds to a diverging family, while immediately after a collision it corresponds to a convergent family.

What happens to the parallel family η_1 ? Since divergent families becomes convergent it is obvious that the parallel family, after reflection, becomes even more convergent. Hence, it will focus before the middle point to the next collision (the point where the family η_3 focuses).

The previous considerations suggest the cone $\mathcal{C}(x, v) = \{\xi \in \mathcal{TM} \mid \alpha\eta_1 + \beta\eta_3 \text{ with } \alpha\beta \geq 0\}$.

²Clearly non smooth perturbations such as the stadium, otherwise the KAM theorem would apply, see Problem 5..

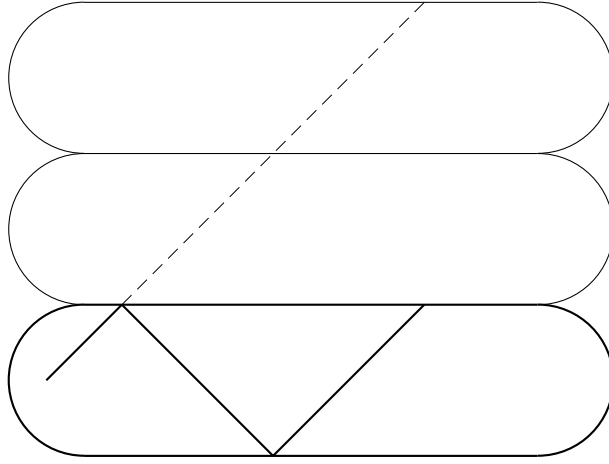


Figure 6.1: Reflecting trajectories in Bunimovich Stadium

Hence, for a trajectory that collides only with a half circle the cone just defined is invariant but not strictly invariant. Since this would be true also for a billiard inside a circle it is clearly not sufficient (the billiard inside a circle has zero Lyapunov exponents, since, as we have already remarked it has an integral of motion). Next, we need to see what happens if a trajectory goes from one circumference to the other (which will happen with probability one). In this case the infinitesimal motion is the same that would happen if the straight line would be not present. In fact, as we have already seen several times, if we reflect the billiard table along the straight lines we can imagine that the motion proceeds in a straight line (see Figure 6.1). Hence the family η_3 will first focus and then diverge for a longer time (and so get closer to the parallel family) than would happen if the collision would be in the same circle. This is exactly what we need to get strict invariance of the cone family.

In conclusion the cone family is strictly invariant each time that the trajectory goes from one half circle to the other. Since this happens almost surely, again we have proven hyperbolicity of the system.

It is interesting to notice that the cone family coincide with the one used in the Sinai Billiard (divergent trajectories) if one looks at it at the right point: the point laying in the intersection between the trajectory before collision and the circle of radius $r/2$ (if r is the radius of the half-circles forming the table) tangent to the the boundary at the next collision with a half-circle (but nowhere else).³

³If the last collision was with a flat wall, then the point is obtained by reflecting the

6.3 Cones

In the previous two sections we have analyzed two special examples using very geometrical arguments. This is very appealing and intuitive, yet, in general, it is useful to have a more algebraic approach. Given the formula for the derivative of the flow obtained in section 5.3, it is very easy to implement the necessary formulae.

As we have already seen, provided we restrict ourself to looking in the right place, we can limit ourselves to considering the standard cone $\mathcal{C}_+ = \{\alpha\eta_1 + \beta\eta_2; \alpha\beta \geq 0\}$.

6.3.1 Examples

6.3.1.a Dispersing (Sinai)

As we have seen, in general dispersing billiards consist in a region $\mathcal{B} \in \mathbb{T}^2$ with strictly convex piecewise smooth boundaries. The cone family in all cases is always $\mathcal{C}_+$. By the results of section 5.3 the map from a collision to the next is given by

$$DT = \begin{pmatrix} 1 & 0 \\ \tau & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{2K}{\sin \varphi} \\ 0 & 1 \end{pmatrix}$$

where $K \geq K_- > 0$ is the curvature of the boundary at the collision point, τ is the time of fly between the two collisions, and φ the collision angle. Clearly $DT\mathcal{C}_+$ is strictly contained in $\mathcal{C}_+$, thus all the Sinai billiards are hyperbolic (since collisions take place almost surely, see Problem 5.13).

6.3.1.b Bunimovich

In this case, as we have already pointed out, the cone to consider is again $\mathcal{C}_+$, but only at special points. The cone family can then be obtained by transporting the cone forward ($d\phi^t\mathcal{C}_+$) until the next point where it is already defined.⁴ The special point is the middle of the chord, that is the point at a distance $\tau = -\frac{\sin \varphi}{K}$ from the next collision; where $K = -r^{-1}$ is the curvature of the interior of the boundary of the half-circles. One can easily check algebraically all the geometric assertion of section 6.2 (but see Problems 6.4 Problem 6.5 to appreciate the advantage of this algebraic formulation).

billiard so the trajectory backward looks straight, determining the point and then reflecting back to find the real point on the trajectory.

⁴Note that this cone family is a bit different from the one defined in ???. Nevertheless, a bit of thought will shows that it is essentially equivalent. The point that I wish to make here is that several cone family can do the job and it is often convenient to consider different ones for different arguments (see ??? for an interesting use of such a principle).

6.3.1.c Semi-dispersing

Although the only possible cone family is constitute again by the divergent trajectories, as in the dispersing (Sinai) case, there is no general theory for the study of hyperbolicity of semi-dispersing billiards, thus their hyperbolicity must be explored case by case. Nevertheless, there are large classes of examples that are very well understood. Here we just mention a couple of cases.

6.3.1.d No straight lines

The easiest case is when $K = 0$ only on a set of measure zero of the boundary. Then, since the only trajectories that are not strictly monotone are the one either never colliding or colliding only with the parts of the boundary with zero curvature, and since such trajectories are of measure zero, the hyperbolicity follows by Wojtkowski theorem.

6.3.1.e Polygonal billiards with strictly convex obstacles

The easiest example is obtained by the Sinai billiard adding reflecting walls to delimit a fundamental domain. In this case we have a motion in a square containing an obstacle (the disk). For such an example the hyperbolicity follows from our previous discussion of the “ergodicity” of the motion in a rectangular box (see Examples 5.0.1–Polygonal Billiards) In particular, we have that for almost all the velocities $v = (v_1, v_2)$ (precisely the ones for which $v_1/v_2 \notin \mathbb{Q}$) the motion is ergodic on the set $\mathcal{B} \times \{(\pm v_1, \pm v_2)\}$. This means that almost all the trajectories will eventually collide with the obstacle whereby becoming strictly monotone, hence the hyperbolicity.

6.4 Geometric representation of cones in two- d

In the previous discussion we have seen that it is particularly important to keep track of the point at which an infinitesimal family focuses. This point of view can be brought to its extreme consequences by noticing that the distance of the focusing point along the trajectory can be used as a projective coordinate in the tangent space. It is in fact clear that all the language of cones is naturally translated in projective terms (the only thing that decides if a vector belongs or not to a cone is its direction and not its length). In formulas, if $\xi = \alpha\eta_1 + \beta\eta_2$ then the focusing point is at the distance $-\frac{\alpha}{\beta}$ (when $\alpha\eta_1 + \beta\eta_2 + \beta t\eta_1 = \beta\eta_2$). In this language the description of the systems becomes very geometric: a point in the tangent bundle becomes simply a point in Q (the point at which the moving point is situated) an oriented line (the direction in which the point is moving) and another point (the point at which the family of trajectories associated to the tangent vector would

focus—note that this point can very well be out of Q). Even more drastically, we can forget about the first point, since different points correspond simply to different times, and the free evolution between collisions does not contain any relevant information. The relevant behavior of the Jacobi fields is then described by the so called “mirror equation:” calling f_0 the distance (measured in the direction of the motions) of the focusing point from the collision point before collision and f_1 the distance after, it holds

$$-\frac{1}{f_0} + \frac{1}{f_1} = \frac{2K}{\sin \varphi}.$$

This allows us to give to the cones a very geometric interpretation (see Figure ??)

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6.5 Convex billiard (two-D)

The representation discussed in the previous section is particularly suited to generalize Bunimovich stadium and obtain other convex billiards with non-zero Lyapunov exponents. Here we will give just an idea of what can be accomplished, for more details see [66], [93], [32].

Consider a sufficiently short $\mathcal{C}^{(2)}$ piece of a convex curve with the following property: each parallel family colliding with it will focus between each two consecutive reflections and focus again, at a distance less than some fixed constant, after leaving; see figure 10 (we will call pieces of curve “focusing arcs”).

By the mirror equation we know that, if (φ, s) is the collision point, the parallel family ($f_0 = \infty$) focuses, after reflection, at the intersection between the trajectory and the circle of radius $\frac{1}{4K}$ tangent to the arc at the reflection point. From the figure it is clear that if such arcs are sufficiently far apart and are joined by straight lines, then there exists the wanted invariant cone family.

So far we have seen that it is possible to use some very intuitive geometric ideas to prove that a given system is hyperbolic. Yet, such systems are certainly not uniformly hyperbolic. To see this consider a trajectory that never experience a collision in a Sinai billiard or a periodic trajectory that never collides with the half circles in the Bunimovich stadium; both trajectories have